A Map of Metaproperties

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Abstract

This note summarizes several algebraic and syntactic properties, as studied in the papers *Amalgamation and interpolation in ordered algebras*[1] and *Amalgamation properties and interpolation theorems for equational theories*[2]. It first recalls the definitions of the deductive interpolation property (DIP) and the flat amalgamation property (FAP) from [1], together with the theorem establishing their equivalence. Then we introduce the equational interpolation property (EIP) from [2], recall the corresponding definition of FAP used there, and state their equivalence. We then rewrite the proof of one direction and refer the reader to [3] for the converse. In this way, the equivalence from [1] is extended by adding EIP as an additional equivalent condition coming from [2], enriching the final diagram of relationships between metalogic properties studied in [1].

1 Notation

We will use mainly the same notation as in the paper [1]: *signature* is a pair $L = \langle L, \tau \rangle$ consisting of a non-empty countable set L of operation symbols and their arities. For $Y \subseteq X$, let $\mathrm{Eq}(Y)$ be the set of equation over variables in Y. $\mathrm{Var}(\Sigma)$ is the set of variables occurring in a set of equations Σ.

Fix a variety $\mathcal{V}$ of algebras. We write $\Sigma \models_{\mathcal{V}} \varepsilon$ for equational consequence in $\mathcal{V}$. For $Y \subseteq X$, let $F_{\mathcal{V}}(Y)$ denote the free algebra of V on Y. Assume that $\mathcal{V}$ admits free products. Then $A * B$ denotes the free product of algebras A, B from V.

Later in Theorem 3.4, T denotes an equational theory and $\mathrm{Mod}(T)$ its class of models. Conversely, given $\mathcal{V}$ we may fix a theory T axiomatizing it, so that $\mathcal{V} = \mathrm{Mod}(T)$.

2 DIP and FAP from [1]

We first recall two properties from *Amalgamation and interpolation in ordered algebras*.

Definition 2.1 (DIP[1])**.** A variety $\mathcal{V}$ has the *deductive interpolation property* (DIP) if for any set of variables Y, whenever

(i) $\Sigma \cup \{\varepsilon\} \subseteq \mathrm{Eq}(Y)$ and $\mathrm{Var}(\Sigma) \cap \mathrm{Var}(\varepsilon) \neq \emptyset$,

(ii) $\Sigma \models_{\mathcal{V}} \varepsilon$,

there exists $\Delta \subseteq \mathrm{Eq}(Y)$ such that

$$\Sigma \models_{\mathcal{V}} \Delta, \qquad \Delta \models_{\mathcal{V}} \varepsilon, \qquad \mathrm{Var}(\Delta) \subseteq \mathrm{Var}(\Sigma) \cap \mathrm{Var}(\varepsilon).$$

Definition 2.2 (FAP[1])**.** Assume that $\mathcal{V}$ admits free products. We say that $\mathcal{V}$ has the *flat amalgamation property* (FAP) if whenever $A, B \in \mathcal{V}$ with $A \leq B$ and $Y \neq \emptyset$, the induced homomorphism

$$A * F(Y) \to B * F(Y)$$

is injective.

Now we recall the theorem regarding the relationship of those properties:

Theorem 2.3 (Thm 23, (2) $\leftrightarrow$ (3) [1])**.** *Let $\mathcal{V}$ be a variety admitting free products. Then the following are equivalent:*

$$\mathcal{V} \textit{ has DIP} \qquad \Longleftrightarrow \qquad \mathcal{V} \textit{ has FAP.}$$

Metcalfe omits the proof in his paper, noting it is a simplification of proof of Thm25[1].

3 EIP and FAP from [2]

We now recall the the second and third equivalent definitions of FAP in Bacsich's paper. The second one makes it straightforward that Bacsich's and Metcalfe's FAP definitions are equivalent, while the third one is used in the proof of Theorem 3.4. Bacsich states these conditions for an arbitrary set X. Here we restrict to non-empty X, as in [1].

Definition 3.1 (FAP, lemma 2.3. (2)[2])**.** Let T be an equational theory, and let X be a non-empty set. Whenever $A \to B$ is an embedding in T, then the natural morphism

$$A * \langle X \rangle \;\to\; B * \langle X \rangle$$

is an embedding.

Definition 3.2 (FAP, lemma 2.3. (3)[2])**.** Let T be an equational theory and let X be a non-empty set. If $A \leq B$ in T, then $A(X)$, the subalgebra of $B * \langle X \rangle$ generated by A and X, is equal to $A * \langle X \rangle$.

Definition 3.3 (EIP, Def. 2.2.[2])**.** T is said to have the *equational interpolation property* (EIP) if whenever $T + \alpha(\bar{x}, \bar{y}) \to \beta(\bar{y}, \bar{z})$ there is $\gamma(\bar{y})$ such that $T + \alpha(\bar{x}, \bar{y}) \to \gamma(\bar{y})$ and $T + \gamma(\bar{y}) \to \beta(\bar{y}, \bar{z})$, where α, β, γ are conjunctions of atomic formulas, and $\bar{x}, \bar{y}, \bar{z}$ are pairwise disjoint lists of variables, with $\bar{y}$ the shared variable list (subject to Bacsich's standing convention concerning an empty shared list).

Theorem 3.4 (Lemma 2.4.[2] and Jónsson's result[3])**.** *T has EIP iff T has FAP.*

Proof. Here $\Delta(A)$ denotes the atomic diagram of A in the language expanded by constants naming the elements of A (and similarly for B). Thus $\bar{a}$ and $\bar{b}$ below are used as parameters. Equivalently, they may be replaced by fresh variables and then instantiated.

First we prove implication EIP $\to$ FAP. Assume that $A \leq B \leq B * \langle X \rangle$. Now we want to show that also $A(X) = A * \langle X \rangle$, which means that for all $\bar{a} \in A$, $\bar{u} \in X$, and atomic $\beta(\bar{x}, \bar{u})$, if $A(X) \vDash \beta(\bar{a}, \bar{u})$ then $T + \Delta(A) \vdash \beta(\bar{a}, \bar{u})$ (by the definition of coproduct and free algebra[4]). Let $A(X) \vDash \beta(\bar{a}, \bar{u})$. Since $A(X) \leq B * \langle X \rangle$, $B * \langle X \rangle \vDash \beta(\bar{a}, \bar{u})$, and so $T + \Delta(B) \vdash \beta(\bar{a}, \bar{u})$: hence there is $\alpha(\bar{x}, \bar{y})$ and $b \in B$ such that $B \vDash \alpha(\bar{a}, \bar{b})$ and $T + \alpha(\bar{a}, \bar{b}) \vdash \beta(\bar{a}, \bar{u})$.

By EIP there is γ such that $T + \alpha(\bar{a}, \bar{b}) \vdash \gamma(\bar{a})$ and $T + \gamma(\bar{a}) \vdash \beta(\bar{a}, \bar{u})$. It follows easily that $A \vDash \gamma(\bar{a})$ and so $T + \Delta(A) \vdash \beta(\bar{a}, \bar{u})$.

The implication FAP $\to$ EIP is a result of Jónssons work[3] $\square$

4 summary diagram

Let $T_{\mathcal{V}}$ be an equational theory axiomatizing $\mathcal{V}$. Identifying $\mathcal{V}$ with the class $\mathrm{Mod}(T_{\mathcal{V}})$ of its models (with homomorphisms as morphisms), and under the standing assumption that $\mathcal{V}$ admits free products, Bacsich's coproduct/free extension $A * \langle X \rangle$ is the same construction as the $\mathcal{V}$-free product $A * F(Y)$ used in [1]. Hence $\mathcal{V}$ has FAP iff $T_{\mathcal{V}}$ has FAP (in Bacsich's sense).

Combining Thm 2.3 and Thm 3.4, we obtain the following equivalence relation:

$$\mathcal{V} \text{ has DIP} \iff \mathcal{V} \text{ has FAP} \iff T_V \text{ has FAP} \iff T_V \text{ has EIP}.$$

Adding this new equivalence to the diragram from fig. 1. from [1], we obtain the following enriched diagram of relationships between metalogical properties studied in [1]:

$$\begin{array}{ccccccc}
\text{FAP} + \text{CEP} & & & & & & \text{EIP} \\
\Updownarrow & & & & & & \Updownarrow \\
\text{TIP} & \Rightarrow & \text{AP} & \Rightarrow & \text{WAP} & \Rightarrow & \text{FAP} \\
\Updownarrow & & \Updownarrow & & \Updownarrow & & \Updownarrow \\
\text{MIP} & \Rightarrow & \text{RP} & \Rightarrow & \text{CDIP} & \Rightarrow & \text{DIP} \\
\Updownarrow & & & & & & \\
\text{DIP} + \text{EP} & & & & & &
\end{array}$$

References

[1] G. Metcalfe, F. Montagna, C. Tsinakis, *Amalgamation and interpolation in ordered algebras*, Journal of Algebra **402** (2014), 21–82.

[2] P. D. Bacsich, *Amalgamation properties and interpolation theorems for equational theories*, Algebra Universalis **5** (1975), 45–55.

[3] B. Jónsson, *Extensions of relational structures*, in *The Theory of Models: Proceedings of the 1963 Symposium at Berkeley*, North-Holland, Amsterdam, 1965.

[4] G. Grätzer, *Universal Algebra*, Van Nostrand, Princeton, 1968.